disclosure, the Commission proposed to require a registrant to describe any analytical tools, such as scenario analysis, that the registrant uses to assess the impact of climate-related risks on its business and consolidated financial statements, and to support the resilience of its strategy and business model in light of foreseeable climate-related risks.[535] The Commission proposed to define scenario analysis to mean a process for identifying and assessing a potential range of outcomes of various possible future climate scenarios, and how climate-related risks may impact a registrant's operations, business strategy, and consolidated financial statements over time.[536] The proposed definition included an example of how registrants might use scenario analysis.[537]

The Commission proposed to require a registrant that uses scenario analysis to assess the resilience of its business strategy to climate-related risks to disclose the scenarios considered (e.g., an increase of no greater than 3 ºC, 2 ºC, or 1.5 ºC above pre-industrial levels), including the parameters, assumptions, and analytical choices, and the projected principal financial impacts on the registrant's business strategy under each scenario. The Commission further proposed that such disclosure should include both qualitative and quantitative information.[538]

b. Comments

Several commenters supported the proposed rule requiring a registrant to describe any analytical tools, such as scenario analysis, that the registrant uses to assess the impact of climate-related risks on its business and consolidated financial statements, and to support the resilience

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535 *See id.*

536 *See id*. More generally, scenario analysis is a process for identifying and assessing a potential range of outcomes of future events under conditions of uncertainty. See, for example, the definition of "scenario analysis" in TCFD, Recommendations of the Task Force on Climate-related Financial Disclosures, Appendix 5.

537 *See* Proposing Release, section II.C.4.

538 *See id.*

of its strategy and business model in light of foreseeable climate-related risks.[539] One commenter stated that scenario analysis has emerged as a key analytical tool for assessing potential climate-related impacts on a company by allowing market participants to understand multiple possible outcomes while still reflecting a realistic level of uncertainty.[540] This commenter further indicated that disclosure of scenario analysis if used would allow investors to review the general models and projections used by the company in its planning and capital allocation strategy, and would greatly assist investors in understanding a firm's resilience and assumptions about the effects of climate change.[541] Another commenter supported the disclosure of scenario analysis if used because of the importance to investors of forward-looking assessments of climate-related risks in understanding the resilience of a company's climate-related strategy.[542]

Some commenters recommended that the Commission require all registrants to provide scenario analysis disclosure in their climate risk reporting, regardless of whether they otherwise

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539 *See, e.g.*, letters from American Institute of CPAs (June 15, 2022) ("AICPA"); AllianceBernstein; Amer. for Fin. Reform, Sunrise Project *et al*.; Bloomberg; CalSTRS; Ceres; CFA; Council of Institutional Advisors (May 19, 2022) ("CII"); Eni SpA; IAC Recommendation; ICGN; ICI; J. McClellan; Morningstar; Norges Bank; NRDC; Paradice Invest. Mgmt.; Member of the U.S. House of Representatives Kathy Castor and 130 other House Members (Jun. 17, 2022) ("U.S. Reps. Castor *et al.*"); San Francisco Employees' Retirement System (June 17, 2022) ("SFERS"); Unilever; Vodafone; and Wellington Mgmt.

540 *See* letter from Amer. for Fin. Reform, Sunrise Project *et al*.

541 *See id*.; *see also* letters from ICI (stating that "[i]nformation about scenario analysis can help investors evaluate the resilience of the company's business strategy in the face of various climate scenarios that could impose potentially different climate-related risks"); and Wellington Mgmt. (stating that "disclosure of a scenario analysis enables investors to assess an issuer's risk management process and whether an issuer is considering different climate risk outcomes in its planning").

542 *See* letter from Bloomberg; *see also* letter from Morningstar (stating that scenario analysis is an important analytical tool in which companies may project their performance and results subject to various changes, including, but not limited to, policy interventions, technological advancement, or environmental and physical challenges, and that such analysis would help investors understand circumstances under which the value of a company could be at risk, and how a company's strategy may—or may not—move it forward toward long-term value creation and sustainability).

use scenario analysis.[543] One such commenter stated that requiring scenario analysis disclosure is essential if a registrant's disclosure of material climate-related risks is to be decision-useful for investors.[544] According to that commenter, because scenario analysis requires a registrant to make assumptions regarding different global temperature increase pathways and various potential pathways of decarbonization involving regulatory, technological, and behavioral responses, investors need to know the assumptions and parameters considered by the registrant in order to understand the registrant's disclosure of likely climate-related impacts.[545] One other commenter stated that, "all else being equal," registrants that conduct strong scenario analyses should have more intrinsic value in the securities they offer than issuers that do not plan sufficiently for climate risk.[546]

One commenter stated that the proposed scenario analysis disclosure requirement struck an appropriate balance by requiring registrants to share any scenario analysis that they are otherwise conducting for their business operations while avoiding imposing a potentially difficult or burdensome requirement on those registrants that have not yet conducted such analysis.[547] Some commenters similarly stated that, due to cost concerns, they could only support a requirement to disclose scenario analysis if it was limited to situations in which a registrant has actually used such analysis in its assessment of climate-related risks.[548] Other commenters supported the proposed scenario analysis disclosure requirement but only if the use

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543 *See, e.g.*, letters from Anthesis; NY St. Comptroller; PRI; and SFERS.

544 *See* letter from SFERS.

545 *See id.*

546 *See* letter from Wellington Mgmt.

547 *See* letter from CII.

548 *See, e.g.*, letters from AICPA; J. McClellan; and Unilever.

of scenario analysis reflected an expected material impact on the registrant's business strategy, financial planning, and capital raising.[549] Still other commenters recommended that the Commission require a registrant that does not currently use scenario analysis to explain why it does not do so to prevent the disclosure requirement from acting as a disincentive to the adoption of scenario analysis.[550]

Several commenters opposed or expressed concerns about the proposed requirement to disclose scenario analysis, if used.[551] Some commenters stated that the proposed requirement could result in the disclosure of confidential business information.[552] Other commenters stated that a scenario analysis disclosure requirement that is not qualified by materiality would act as a disincentive to the use of scenario analysis as a climate-related tool.[553] Still other commenters opposed the proposed disclosure requirement because it was too prescriptive and would be costly and burdensome to fulfill.[554] Because of the above concerns, some commenters stated that the disclosure of scenario analysis should be voluntary.[555] Other commenters stated that the required scenario analysis disclosure should be limited to high level trends or material drivers and impacts, and should not cover more detailed parameters, assumptions, and analytical choices

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549 *See, e.g.*, letters from ABA; and AllianceBernstein.

550 *See, e.g.*, letters from BlackRock; ICI; NEI Investments (June 17, 2022) ("NEI"); and NY City Comptroller.

551 *See, e.g.*, letters from Alphabet *et al.*; Amazon; Amer. Bankers; AFPM; CEMEX; Chamber; Chevron; Citigroup; Hydro One Limited (June 16, 2022) ("Hydro One"); Institute of International Finance (June 17, 2022) ("IIF"); NAM; Northern Trust; RILA; Shearman Sterling; Soc. Corp. Gov.; Sullivan Cromwell; the Travelers Companies (June 17, 2022) ("Travelers"); and Western Midstream.

552 *See, e.g.*, letters from AFPM; Amazon; Amer. Bankers; Chevron; Citigroup; GPA Midstream; IIF; NAM; RILA; Shearman Sterling; Soc. Corp. Gov.; Sullivan Cromwell; and Travelers.

553 *See, e.g.*, letters from Chamber; PGIM; Sullivan Cromwell; United Parcel Service, Inc. (Jun. 14, 2022) ("UPS"); and Western Midstream; *see also* letter from Beller *et al*. (opposing a mandatory scenario analysis disclosure requirement because it would stifle innovation).

554 *See, e.g.*, letters from Amer. Bankers; Dimensional Fund; NAM; and Soc. Corp. Gov.

555 *See, e.g.*, letters from Alphabet *et al.*; Beller *et al.*; Chamber; Hydro One; and Northern Trust.

underlying the scenario analysis, as proposed.[556] One commenter stated that scenario analysis disclosure should only be required when it is broadly used by senior management and the board as part of their strategic planning process and when integrated and material to a publicly announced climate-related strategy or initiative.[557]

Some commenters recommended that the Commission require the use of certain publicly available scenario models, such as those published by the Intergovernmental Panel on Climate Change ("IPCC"), the International Energy Agency ("IEA"), or the Network of Central Banks and Supervisors for Greening the Financial System ("NGFS"), to enhance the comparability of the scenario analysis disclosure.[558] Other commenters stated that it should be up to each registrant to choose those scenarios that best fit its particular business or industry and tailor its disclosure accordingly.[559]

c. Final Rule

We are adopting a final rule (Item 1502(f)) requiring the disclosure of scenario analysis under certain circumstances. The disclosure of a registrant's use of scenario analysis can provide important forward-looking information to help investors evaluate the resilience of the registrant's strategy under various climate-related circumstances.[560] Scenario analysis has increasingly been recognized as an important analytical tool in assessing a company's climate-related risk

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556 *See, e.g.*, letters from ABA; and Chevron.

557 *See* letter from Amazon.

558 *See, e.g.*, letters from Anthesis; Bloomberg; CalSTRS; Chevron; and Shell plc (June 17, 2022) ("Shell").

559 *See, e.g.*, letters from American Council of Life Insurers (June 17, 2022) ("ACLI"); J. Herron; and TotalEnergies.

560 *See supra* notes 540-542 and accompanying text.

exposure,[561] and investors have increasingly sought information from registrants about their use of scenario analysis and expressed a need for improved disclosure about such use.[562]

Although some commenters recommended that we require all registrants to include scenario analysis disclosure in their climate risk reporting,[563] we recognize that not every registrant conducts scenario analysis and, as noted above, this rulemaking does not seek to prescribe any particular tools, strategies, or practices with respect to climate-related risks but rather, when material, to provide investors with the information they need to evaluate the climate-related risks faced by the registrant and their potential impacts on the registrant's business, results of operations, or financial condition. Therefore, similar to the proposed rule, the final rule's scenario analysis disclosure requirement will depend on whether and how a registrant uses such analysis. Importantly, the rule will not require any registrant to conduct scenario analysis.

We are, however, adopting modifications in the final rules. For example, we have added a materiality qualifier regarding the disclosure of scenario analysis to address commenters' concern that the proposed requirement could result in disclosure of immaterial information that

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561 *See, e.g.*, letter from AllianceBernstein (stating that "[s]cenario analysis is particularly important for those registrants in emissions-intensive industries where such analysis can demonstrate the quality of impairment testing and increase confidence in asset values"). The Federal Reserve Board's climate scenario analysis pilot program, in which six of the nation's largest banks are voluntarily participating, further demonstrates the increased recognition of scenario analysis as an important tool to assess climate-related financial risks. *See* Board of Governors of the Federal Reserve System, *Federal Reserve Board announces that six of the nation's largest banks will participate in a pilot climate scenario analysis exercise designed to enhance the ability of supervisors and firms to measure and manage climate-related financial risks* (Sept. 29, 2022), available at https://www.federalreserve.gov/newsevents/pressreleases/other20220929a.htm.

562 *See, e.g.*, letters from AllianceBernstein (stating that "[w]hile many registrants claim to perform scenario analysis, however, there is little disclosure around assumptions used in these models and how registrants use results impact strategy, business and capital allocation decisions, making their results challenging to compare"); and Ceres (citing evidence from the Climate Action 100+ Benchmark that companies' "scenario analyses leave much room for improvement").

563 *See supra* note 543 and accompanying text.

would be burdensome and costly to produce.[564] We also note that, as with transition plan and use of internal carbon price disclosure, a registrant's scenario analysis disclosure will be subject to a safe harbor.[565] The final rule provides that, if a registrant uses scenario analysis[566] to assess the impact of climate-related risks on its business, results of operations, or financial condition, and if, based on the results of scenario analysis, a registrant determines that a climate-related risk is reasonably likely to have a *material* impact on its business, results of operations, or financial condition, then the registrant must describe each such scenario,[567] including a brief description of the parameters, assumptions, and analytical choices used, as well as the expected material impacts, including financial impacts, on the registrant under each such scenario.[568] We are adopting this disclosure requirement because, if a registrant has used scenario analysis to assess and manage a material climate-related risk, investors need to understand how it conducted that analysis in order to evaluate the registrant's conclusions regarding material impacts on its business, results of operations, or financial condition.

We also have streamlined the proposed scenario analysis disclosure requirements to reduce redundancy in the final rules. For example, we have eliminated the introductory

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564 *See supra* note 554 and accompanying text.

565 *See infra* section II.J.3.

566 We are largely adopting the definition of scenario analysis, as proposed. *See* 17 CFR 229.1500 ("Scenario analysis means a process for identifying and assessing a potential range of outcomes of various possible future climate scenarios, and how climate-related risks may impact a registrant's business strategy, results of operations, and financial condition over time.") We have deleted from the definition the example that "registrants might use scenario analysis to test the resilience of their strategies under certain future climate scenarios, such as those that assume global temperature increases of 3 °C, 2 °C, and 1.5 ºC above pre-industrial levels" because we do not wish to convey the impression that these scenarios are required should a registrant elect to conduct scenario analysis.

567 *See* 17 CFR 229.1502(f). Conversely, if a registrant conducts scenario analysis and determines from its results that it is not likely to be materially impacted by a climate-related risk, no disclosure about its use of scenario analysis is required under Item 1502(f).

568 *See id.*

provision in the rule proposal requiring a registrant to describe the resilience of its business strategy in light of potential future changes in climate-related risks. Because companies use scenario analysis to test the resilience of their business strategies under varying future climate scenarios, and because such use is explained in the definition of scenario analysis (in Item 1500) that we are adopting largely as proposed,[569] if registrants are required to disclose their use of scenario analysis under the final rules, such disclosure likely would include a description of the resilience of their strategies under various climate scenarios.

The rule proposal would have required a registrant to disclose "any analytical tools, such as scenario analysis" that it uses to assess the impact of climate-related risks on its business. In a modification of the proposed rule, we have eliminated the reference to "any analytical tools" to clarify that the disclosure required by this provision should concern the registrant's use of scenario analysis rather than any other analytical tools. We note that the TCFD's guidance discusses scenario analysis as the primary tool to help companies assess the impacts of climate-related risks on their business strategies, and therefore this change should eliminate any confusion about what other analytical tools might fall under the scope of the requirements.[570]

In another change from the rule proposal, we have added the term "brief" to modify the "description of the parameters, assumptions, and analytical choices used" prong of the scenario analysis disclosure provision. The adopted provision will continue to elicit disclosure that will enhance investors' assessment of the resiliency of a registrant's strategy while also mitigating the compliance burden for registrants. Requiring a *brief* description of the parameters, assumptions, and analytical choices used, together with a description of the projected *material* financial

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569 *See* 17 CFR 229.1500.

570 *See* TCFD, *supra* note 332332.

impacts on the registrant's business strategy under each scenario, should help elicit disclosure that neither burdens investors with immaterial detail nor unduly adds to a registrant's compliance burden. As with disclosure related to transition plans, we reiterate that our focus in adopting these requirements is neither on incentivizing nor disincentivizing any particular risk management practice but rather on providing investors with the information they need with respect to the particular practices of a registrant in order to make informed investment and voting decisions.

These revisions to the proposed rule also address commenters' concern that the required scenario analysis disclosure could result in the disclosure of confidential business information.[571] If a registrant has used scenario analysis to determine that an identified climate-related risk is likely to have a material impact on its business, results of operations, or financial condition, it is important for investors to receive disclosure about that material impact. The registrant will not, however, be required to provide a lengthy description of the underlying parameters and assumptions that may be more likely to reveal confidential business information.

Although some commenters recommended that we require the use of one or more climate scenario models,[572] the final rules do not impose any specific risk management model. By requiring disclosure based on whether a registrant has determined to conduct scenario analysis as part of its consideration of material climate-related risks, a registrant will be able to select the climate scenario model or models that it believes best fits its particular industry or business, or its climate risk assessment approach. This approach will provide useful information to investors

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571 *See supra* note 552 and accompanying text.

572 *See supra* note 558 and accompanying text.

about the resilience of a registrant's climate-related business strategy while also helping to limit the registrant's compliance burden relating to scenario analysis disclosure under the final rules.

The proposed scenario analysis disclosure provision would have included as an example of potential scenarios to be considered "an increase of no greater than 3 ºC, 2 ºC, or 1.5 ºC above pre-industrial levels."[573] Because this was for illustrative purposes only, and because we have removed the same example from the definition of scenario analysis to avoid conveying the impression that these scenarios are required,[574] we have also removed the example from Item 1502(f).

To further streamline the scenario analysis disclosure requirement, we have removed the proposed provision stating that the disclosure should include both qualitative and quantitative information.[575] We recognize that, as noted by some commenters, scenario analysis practices are still evolving,[576] and that, in the early stages of use, a registrant's disclosure regarding its use of scenario analysis may be qualitative. As a registrant's use of scenario analysis becomes more sophisticated, we would expect its disclosure of the results of scenario analysis to become more quantitative, particularly when discussing the expected material financial impacts on the registrant's business strategy, under each considered scenario, which, like the proposed rule, must be addressed should a registrant be required to disclose its use of scenario analysis. Streamlining the proposed scenario analysis disclosure requirement in this way will enable a registrant to determine the mix of qualitative and quantitative disclosure that best fits its particular circumstances when satisfying its obligations under the final rule.

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573 *See* Proposing Release, section II.C.4.

574 *See supra* note 566 and accompanying text.

575 *See id.*

576 *See, e.g.*, letters from Bloomberg; and Chamber.